



João Filipe Santos Lopes

Full-Stack AI Developer | Embedded Systems | LLM Integration

joao.santos.lopes@tecnico.ulisboa.pt | 910 567 809 | Lisbon, PT

[LinkedIn](#) [GitHub](#) [itch.io](#)

★ Selected to present at the MIT Portugal Innovation Workshop — Boston, USA (2026)

Profile

MSc Electrical & Computer Engineering graduate (IST, Lisbon) with hands-on experience building production AI pipelines, agentic RAG architectures, and full-stack applications. Works across the stack from embedded/edge AI to enterprise AI architecture, with shipped projects in both areas.

Technical Skills

Backend / APIs	FastAPI • Python • REST APIs • Docker • Expo • GO
LLM / AI Stack	LangChain • Ollama • RAG • Amazon Bedrock • OpenClaw / NemoClaw • N8N
Frontend	Next.js • JavaScript / TypeScript • React
Embedded / Hardware	C/C++ • STM32 • VHDL • FPGA (Vitis HLS / Vivado) • MatLab
Other	Git / GitHub • Unity / Godot • 3D Modeling • Office Suite

Professional Experience

AI Developer Intern — OGMA – Aircraft Maintenance

2025 – 2026

- Designed and deployed AI solution architectures meeting production engineering requirements.
- Built automated AI-generated content pipelines for enterprise applications and internal documents.
- Developed agentic AI workflows integrating LLMs with existing company systems.

Technical Support Engineer — Instituto Superior Técnico (IST)

2024 – 2025

- IT support and troubleshooting for students and academic staff across the university.

Game Development Teacher — St. Julian's School, Carcavelos

2023 – 2025

- Taught Godot programming, 3D modeling, and robotics (all classes in English) to 35 students aged 5–15.

Programming Teacher — Sharkcoders Almada

2022 – 2024

- Taught Python, JavaScript, Scratch, 3D modeling, and game development to 6 classes of ~30 students (ages 12–17).

Projects

Agentic AI Architectures — Personal / OGMA

2024 – 2025

- Designed local AI agent systems using RAG with Ollama, LangChain, OpenClaw/NemoClaw, and N8N.
- Automated enterprise document generation pipelines combining multiple LLM calls via orchestration graphs.
- Deployed containerised services with Docker for reproducible, scalable agentic workflows.

AI Accelerator on SoC-FPGA (Master Thesis) — IST

2025

- Implemented hardware accelerators for attention mechanisms and image preprocessing on a Xilinx SoC-FPGA.
- Built a hybrid hardware-software system leveraging the ARM processor for the software stack while offloading compute-heavy operations to custom HLS IP cores.

- Reduced execution cycles ~40× through parallel accumulator architectures and 8-bit quantization.

Embedded AI Deployment Tool — IST / Personal

2024

- Built a compiler toolchain for rapid deployment and optimisation of AI neural networks on low-power microcontrollers (ESP32, STM32).
- Applied quantization and graph optimisation to fit and accelerate models within tight memory and power budgets.

Moonitor RFID – Precision Livestock Farming — IST

2023

- Developed a remote cow-welfare monitoring product using RFID tags, embedded firmware, and a data backend.

Education

MSc Electrical and Computer Engineering — Instituto Superior Técnico, Lisbon

2023 – 2025

- Specialisation: Electronic Circuits & Systems; secondary area: Control, Robotics & AI. Average: 17/20.
- Academic Merit Diploma 2022/2023 — awarded to the top students in the programme.
- Thesis: Hardware-Software AI Accelerator for Hyperspectral Image Target Detection on SoC-FPGA.
- Selected to present the AI accelerator project at the MIT Portugal Innovation Workshop, Boston, USA (2026).

BSc Electrical and Computer Engineering — Instituto Superior Técnico, Lisbon

2020 – 2024

Extra-Curricular

Member & Partnerships Leader — GameDev Técnico (IST Student Group)

2020 – 2025

- Shipped 14+ published games with the student group, several developed for international competitions.
- As Partnerships Leader (2023–2025), secured sponsorships and organised events with 30+ participants and industry guests.

Additional

- Technical support volunteer — 73rd EAAP Annual Meeting (2022), international event with 1500+ participants.
- European exchange on rural resilience — Stuttgart, Germany (2023).
- Collaborator — Ruralidade Verde family enterprise (2021 – present).